

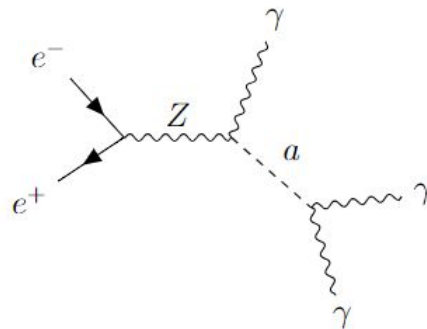
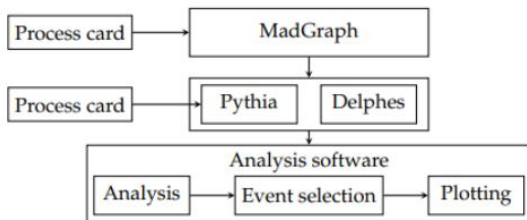
Simulating and Investigating Long-Lived Axion-Like Particles at the Proposed FCC-ee

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Overview

- Continuing on work done searching for ALP signatures at the FCC-ee Z pole run: integrated luminosity $L = 150 \text{ ab}^{-1}$, center-of-mass energy $\sqrt{s} = 91.0 \text{ GeV}$
- Simulation of data (100000 events) for signal and background processes using MadGraph 3.5.4 (importing ALP_NLO_UFO model), Pythia 8, Delphes
- Analysis, selections, and plotting done within FCCAnalyses framework
- An investigation into ALP lifetimes is carried out, useful selections are suggested, and a sensitivity analysis of the Z pole run of the FCC-ee is done
- Signal process:** $e^+e^- \rightarrow Z \rightarrow \gamma, \text{ALP}$ ($\text{ALP} \rightarrow \gamma\gamma$)





Lifetime Analysis

Preexisting ALP Lifetime Theory

- Decay width depends on ALP mass and coupling constant:

$$\Gamma(a \rightarrow \gamma\gamma) = \frac{4\pi\alpha^2 m_a^3}{\Lambda^2} |C_{\gamma\gamma}^{\text{eff}}|^2,$$

where lifetime $\tau(a \rightarrow \gamma\gamma) = \frac{\hbar}{\Gamma(a \rightarrow \gamma\gamma)}$ and $\Lambda = 32\pi^2 f_a |C_{GG}^{\text{eff}}|$

- C_{GG}^{eff} is set to 0.5, $C_{\gamma\gamma}^{\text{eff}}$ is simplified to $C_{\gamma\gamma}$, and:
 - f_a is updated from prior searches of this signal process from 1000 GeV to $\frac{1000}{16\pi^2} \text{ GeV} \approx 6.33 \text{ GeV}$
- Combining all equations to solve for ALP lifetime τ yields:

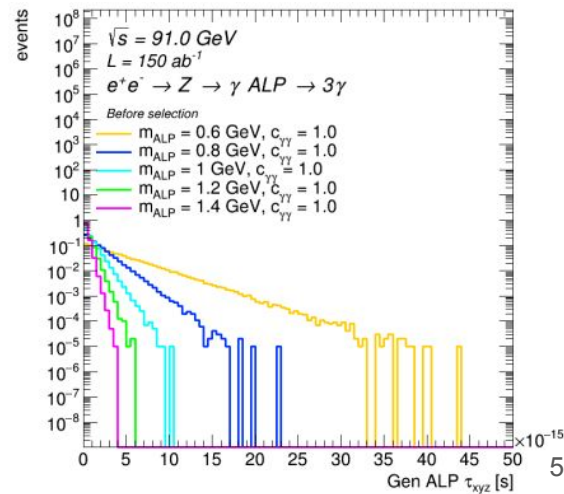
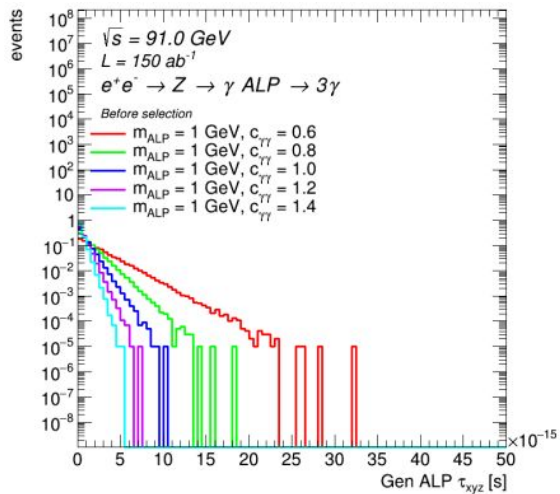
$$\tau(a \rightarrow \gamma\gamma) = \frac{64\pi^3 \hbar f_a^2}{\alpha^2 m_a^3 C_{\gamma\gamma}^2}$$

Lifetime from Simulations

- Within "stage 1" in the FCCAnalyses framework, ALP lifetime at the "gen" level is calculated from the decay vertices of the ALP:

$$L_{xyz} = \sqrt{v_x^2 + v_y^2 + v_z^2}, \quad \tau_{xyz} = \frac{L_{xyz} m_a}{10^3 p_a c}$$

- where the m_a/p_a term accounts for a change in reference frames: τ is the ALP lifetime in the inertial reference frame
- ALP lifetime plots:
 - As predicted from the preexisting theory, τ **decreases** with both an **increase in coupling constant** and an **increase in mass**



Confirmation: Simulation Matches Theory

- As expected, the theoretically computed lifetime values are within the margin of error of the Monte Carlo generated values

m_a (GeV)	$c_{\gamma\gamma}$	Analytically calculated τ_{xyz} (s) (7)	Mean of MC generated τ_{xyz} graph (s)
1.0	0.6	2.378E-15	2.373E-15 $\pm$ 7.542E-18
1.0	0.8	1.338E-15	1.338E-15 $\pm$ 4.250E-18
1.0	1.0	8.561E-16	8.561E-16 $\pm$ 2.712E-18
1.0	1.2	5.945E-16	5.944E-16 $\pm$ 1.885E-18
1.0	1.4	4.368E-16	4.359E-16 $\pm$ 1.385E-18
0.6	1.0	3.964E-15	3.948E-15 $\pm$ 1.245E-17
0.8	1.0	1.672E-15	1.670E-15 $\pm$ 5.304E-18
1.0	1.0	8.561E-16	8.561E-16 $\pm$ 2.712E-18
1.2	1.0	4.954E-16	4.960E-16 $\pm$ 1.577E-18
1.4	1.0	3.120E-16	3.120E-16 $\pm$ 9.924E-19

Table 1: Lifetime values calculated from the analytical computation vs. from Monte Carlo generation.



Useful Selections

Background Processes of Interest

- The following background processes were considered, as they have the potential to have similar signatures to the signal process:

Di-photon production: $e^+e^- \rightarrow \gamma\gamma$

Tri-photon production: $e^+e^- \rightarrow \gamma\gamma\gamma$

Quad-photon production: $e^+e^- \rightarrow \gamma\gamma\gamma\gamma$

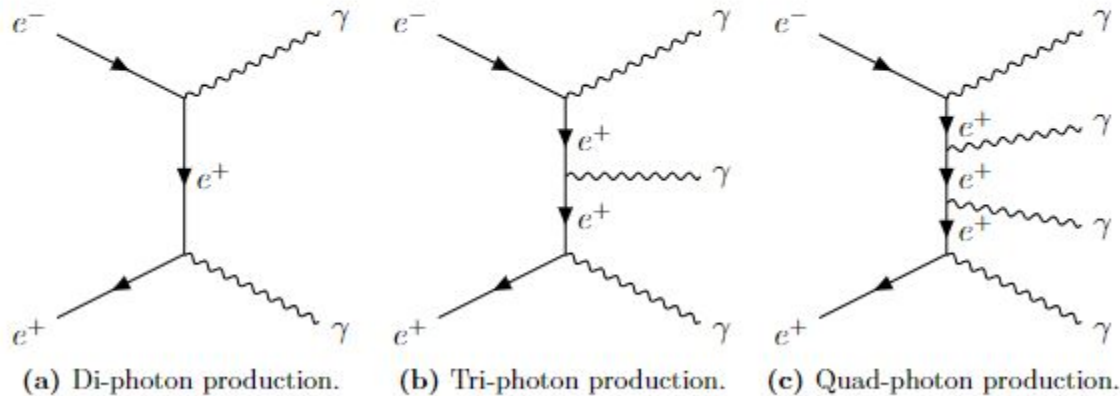
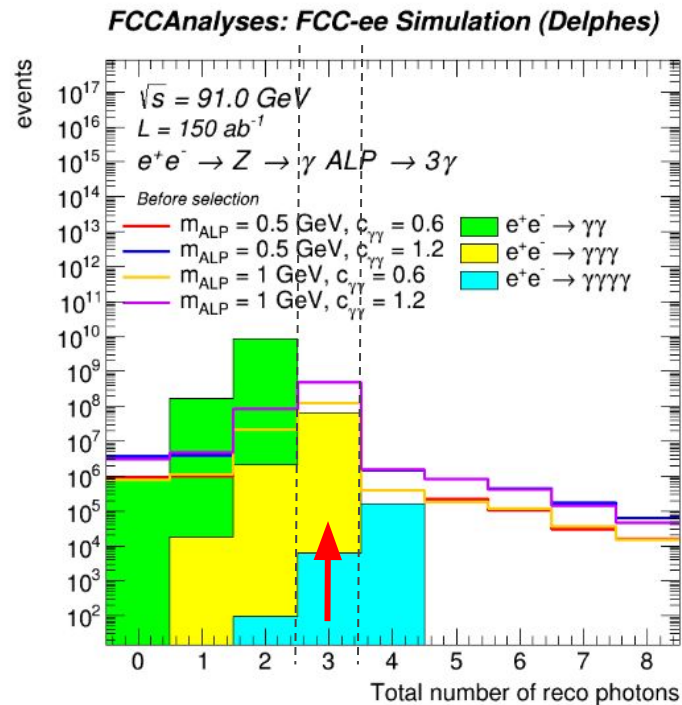


Figure 3: Feynman diagrams for background processes.

Selection: "reco" Photons

- Based on the plot to the right, the first logical selection made was to limit the number of reconstructed photons to **exactly 3 "reco" photons**
- This selection entirely eliminates the di-photon production background and a large amount of the quad-photon production background
 - Further selections necessary with the aim of limiting the amount of the tri-photon production background

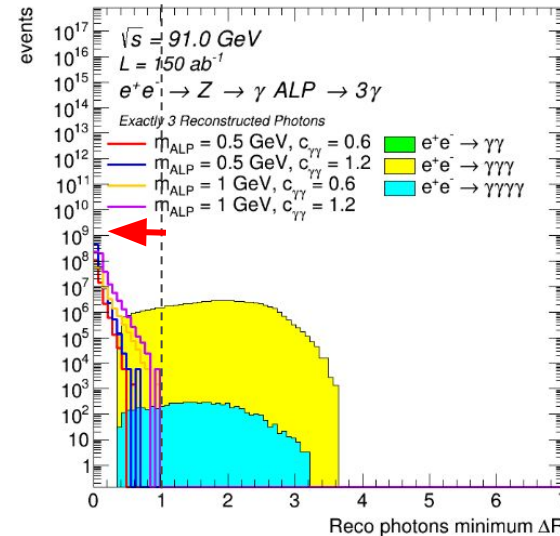
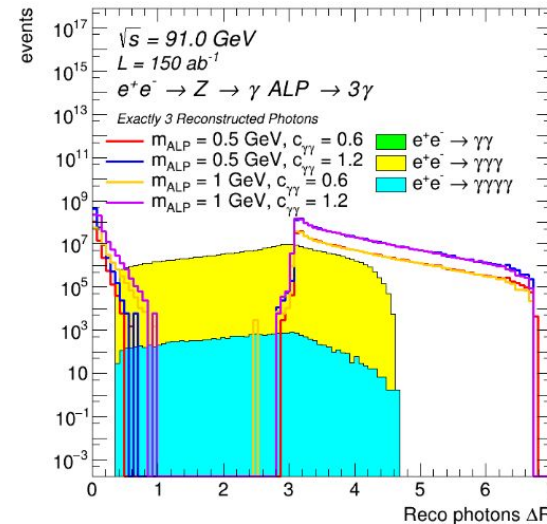


Plots with background normalized to expected number of events assuming integrated luminosity of 150 ab^{-1}

(b) Number of "reco" level photons.

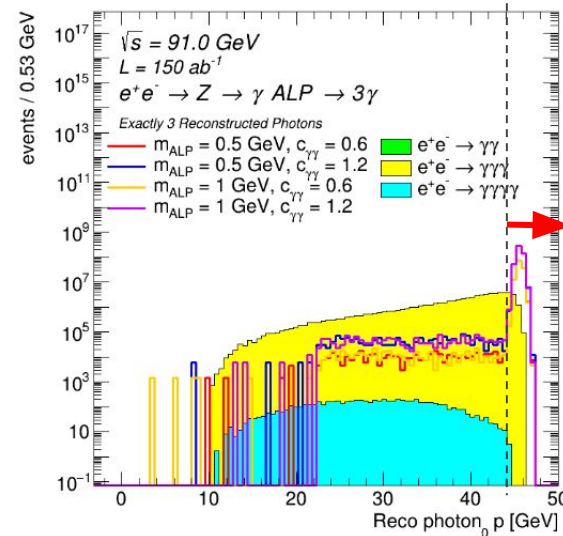
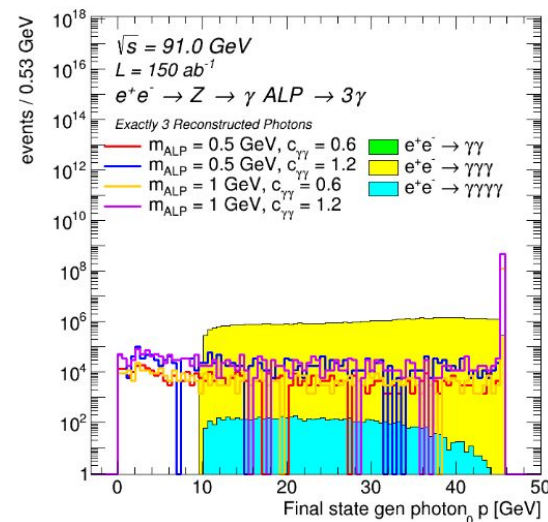
Selection: $\min(\Delta R)$

- Defining ΔR in "stage 1": $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$
 - Represents the angle between two particles, which would be useful because the angle between the 2 photons which decayed from the ALP should often be smaller than the angles between photons in the $ee \rightarrow \gamma\gamma$ background
 - Also added $\min(\Delta R)$, returning the minimum ΔR between any two particles in a certain event (in this case, photons)
- A selection was made to **require $\min(\Delta R)$ to be less than 1.0** in order to eliminate the most background while keeping as much signal as possible



Selection: Leading Photon Momentum

- When considering the signal process, at a center-of-mass energy of $\sqrt{s} = 91.0$ GeV, as the Z boson decays, both the resulting ALP and photon (leading photon) should have the same momentum ($\frac{1}{2} * 91.0 = 45.5$ GeV)
- Selection made to require the **leading photon's momentum, at the "reco" level, to > 44 GeV**



Summary of Selections

		No sel	3 reco photons		$\Delta R < 1$		$p > 44 \text{ GeV}$		Combined	
Signal										
m_a	$c_{\gamma\gamma}$	events	events	eff	events	eff	events	eff	events	eff
0.5	0.6	100000	84089	15.91%	84089	15.91%	83806	16.19%	83806	16.19%
0.5	1.2	100000	83843	16.16%	83843	16.16%	83539	16.46%	83539	16.46%
1.0	0.6	100000	83496	16.50%	83496	16.50%	83190	16.81%	83190	16.81%
1.0	1.2	100000	83557	16.44%	83557	16.44%	83254	16.75%	83254	16.75%
Background										
$ee \rightarrow \gamma\gamma$		100000	0	100.00%	0	100.00%	0	100.00%	0	100.00%
$ee \rightarrow \gamma\gamma\gamma$		100000	97000	3.00%	13009	86.99%	13532	86.47%	10344	89.66%
$ee \rightarrow \gamma\gamma\gamma\gamma$		100000	3868	96.13%	832	99.17%	4	100.00%	3	100.00%

Table 2: The number of events, for a range of signals and backgrounds, both before and after selections have been made. The “ $\Delta R < 1$ ” and “ $p > 44 \text{ GeV}$ ” columns include the default “3 reco photons” selection, and the “Combined” column combines all three previous selections. “No sel” is simply giving the number of events before any selections were made, and “eff” represents the percentage of either signal or background which was removed after the respective selection was made.



Sensitivity Analysis

Normalization

- Events were normalized to the number of expected events:

$$n = \frac{k\epsilon\sigma\mathcal{L}}{N_{tot}} n_{raw} ,$$

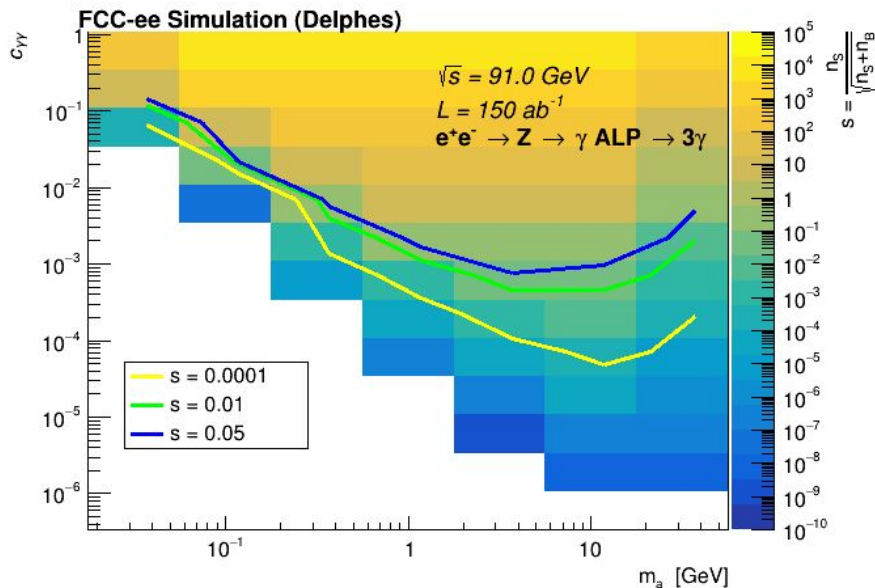
- where k and ϵ are fixed at 1.0
 - Now we can expand the range of ALP masses and coupling constants and study sensitivity of the FCC-ee Z pole run to the signal process
- Sensitivity calculated by taking into account expected number of events for both signal and background:

$$s = \frac{n_S}{\sqrt{n_S + n_B}}$$

- Events (10000 for each ALP mass and coupling constant) were generated for a much larger range of ALP masses and coupling constants

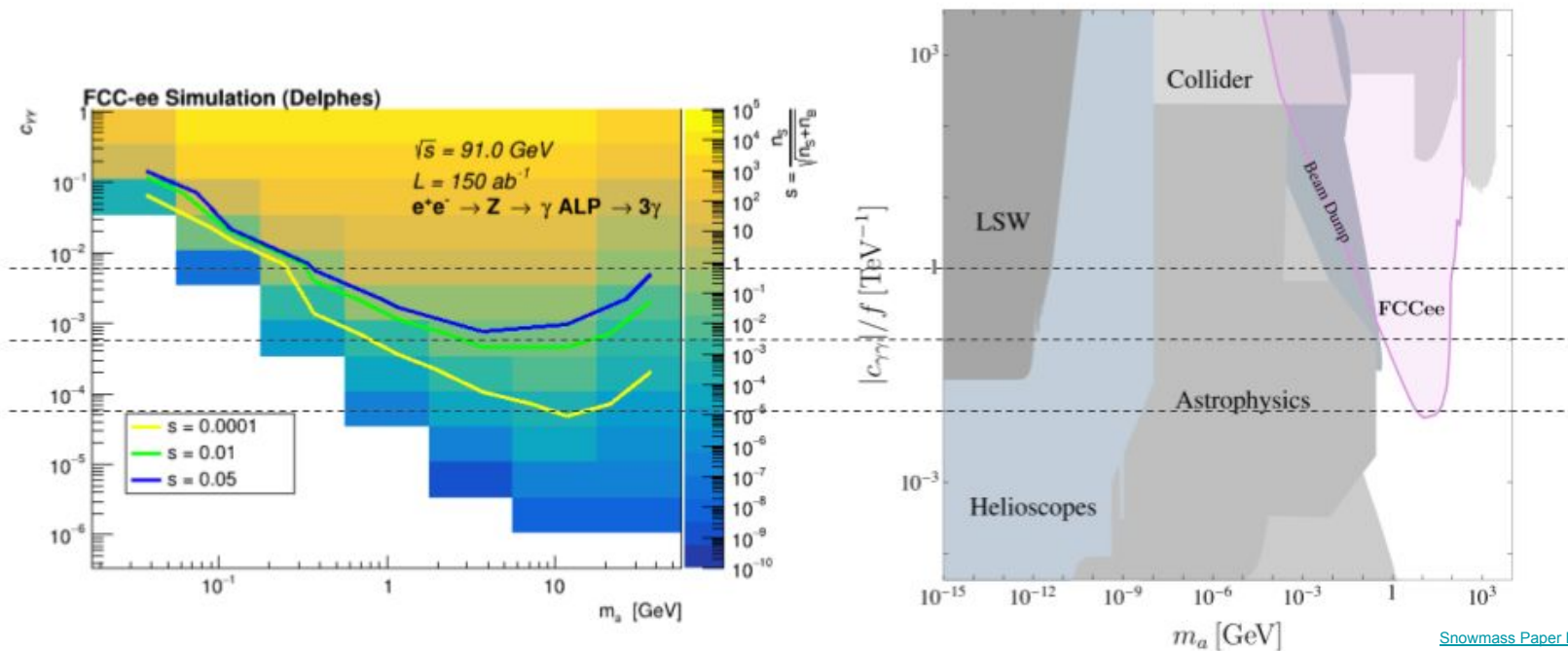
Sensitivity Plot

- Sensitivity was plotted logarithmically along a large range of ALP masses and coupling constants
 - Sensitivity **increases** with an **increase** in coupling constant

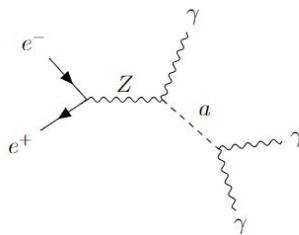


Comparing Sensitivity to Theoretical Prediction

- General shape matches, and the theoretical prediction curve approximately matches a simulated sensitivity of > 0.0001



Conclusion



- ALP production for this signal process were analyzed at the Z pole run of the FCC-ee
 - Many new samples were generated using updated parameters
 - Lifetimes were investigated, and theoretical computations were shown to match Monte Carlo simulation
 - Backgrounds with similar signatures were considered and selections were suggested in order to remove as much of these relevant backgrounds as possible while keeping the majority of the signal
 - A sensitivity analysis was carried out to determine the sensitivity of the FCC-ee Z pole run to a broad range of ALP masses and coupling constants, and the simulated sensitivity was shown to match previous theoretical predictions of sensitivity to this process
- Future of this project: consider calorimeter timing variables at the "reco" level, broaden analysis to look at other potential ALP production processes



Backup

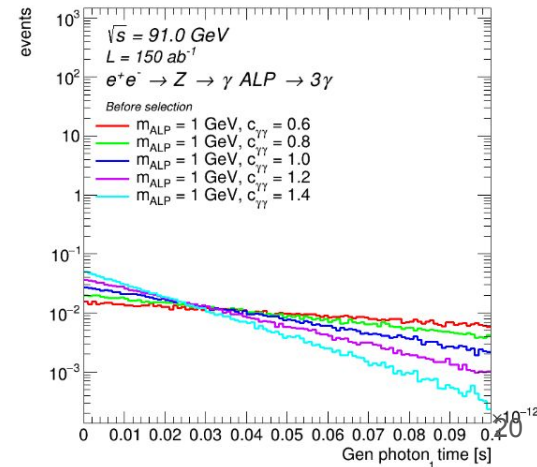
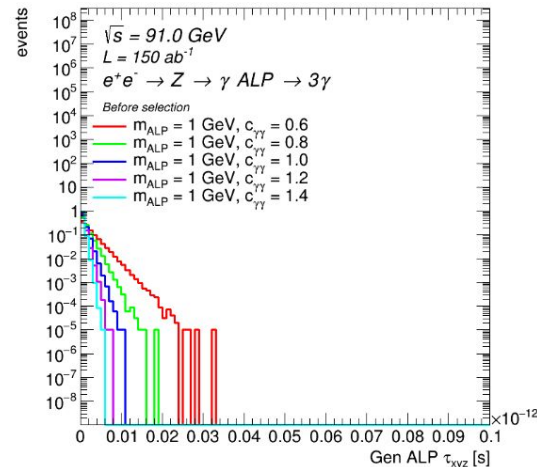
Effective Coupling Constant to Photons

$$C_{\gamma\gamma}^{\text{eff}}(m_a \gg \Lambda_{QCD}) = C_{\gamma\gamma} + \sum_f \frac{N_c^f Q_f^2}{16\pi^2} c_{ff} B_1(\tau_f) + \frac{2\alpha}{\pi} \frac{C_{WW}}{s_w^2} B_2(\tau_W)$$

- c_{ff} and C_{WW} are set to 0 in all simulations done for this analysis, meaning the second two terms go to 0 and the effective coupling constant is just represented as $C_{\gamma\gamma}$

Special Relativity and Reference Frames

- A new variable, photon1_time, was introduced which turned out to represent the non-inertial reference frame, meaning τ represents the inertial reference frame and the following equation $\text{photon1_time} = \gamma \tau_{xyz} = \frac{p_a}{m_a} \tau_{xyz}$
- Indeed, the plots to the right are displaced by exactly a factor of 45.5, or p_a/m_a (45.5 GeV / 1.0 GeV)



Theory/Simulation Discrepancy

- At the point where $m_a = 0.6$ GeV and $c_{\gamma\gamma} = 1.0$, the analytically calculated lifetime value does not fit within the error of the Monte Carlo generated value
 - This is likely due to the long ALP lifetime at that point, causing inconsistency in measurement and potentially causing some ALPs to escape the detector, skewing the Monte Carlo data

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1.0	0.6	2.378E-15	$2.373\text{E-}15 \pm 7.542\text{E-}18$
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1.4	1.0	3.120E-16	$3.120\text{E-}16 \pm 9.924\text{E-}19$

ΔR and Other Angles Definition

- ΔR represents the angles between two particles, or $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$

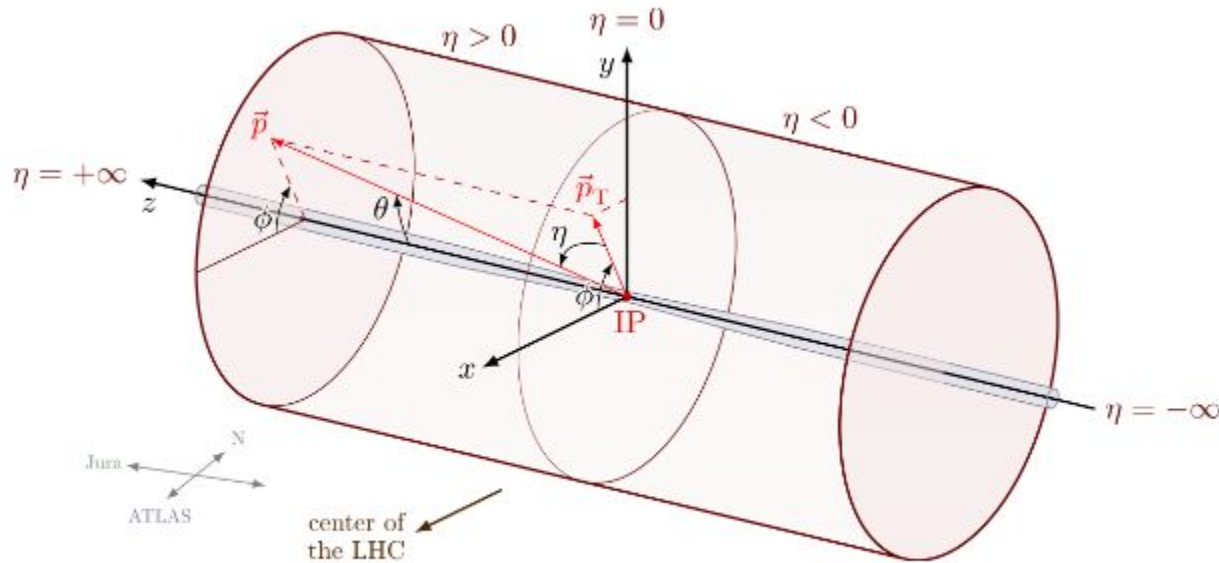
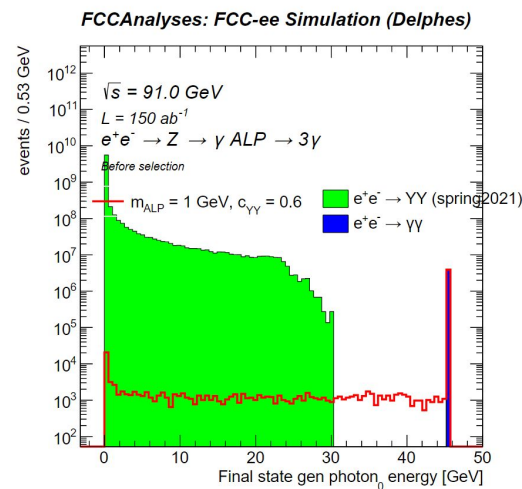
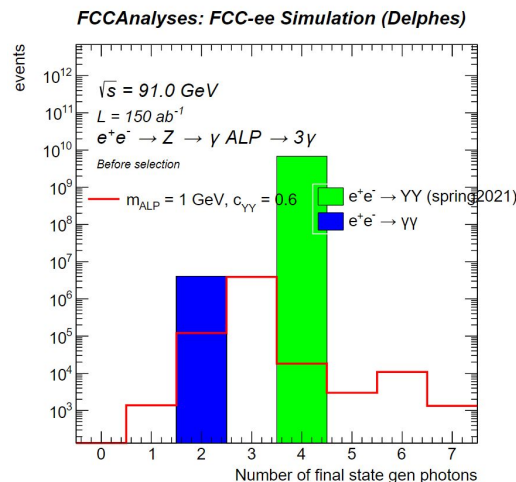


Figure 13: Diagram showing how the angles η and ϕ are defined [18].

Discrepancy between spring2021 $ee \rightarrow \gamma\gamma$ generation (Whizard) and current Madgraph $ee \rightarrow \gamma\gamma$ generation

- When I generate $ee \rightarrow \gamma\gamma$ process using MadGraph, the resulting graphs look very different from those contained in the spring2021 generated file `wzp6_ee_gammagamma_ecm91.root`
 - Notably, the spring2021 files show 4 final state photons, whereas mine shows 2 (which should be correct)





Report

https://indico.desy.de/event/46018/contributions/173511/attachments/91780/124063/Merlin_Summer_Report_ALPs_at_the_FCCee.pdf